Zhenghe Guo

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Academic Website Research Statement GitHub Google Scholar

Final-year undergraduate researcher working on embodied systems and spatial intelligence, with a focus on learned state representations, world modeling, and planning.

Education

Zhejiang University Hangzhou, China
B.Eng. in Agricultural Engineering; Minor in Advanced Class of Engineering Education *2023–2027*
GPA: 3.86/4.00. Undergraduate research with Prof. Mingchuan Zhou.
Selected coursework (scores /100): Fundamentals of Artificial Intelligence (95), Automatic Control Theory (94), Cybernetics (94), Bioproduction Robotics (94), Probability and Mathematical Statistics (93), Frontiers of Artificial Intelligence (93), Theoretical Mechanics (93), Heat Transfer and Fluid Mechanics (92), Robotics Technology and Practice (91), Fundamentals of Mechanical Manufacturing (91), Mechanics of Materials (90).

National University of Singapore Singapore
Visiting Student, Statistics and Data Science *Jun–Jul 2025*
GPA: 4.00/4.00; decision-tree methods instructed by Prof. Wei-Yin Loh.

Research Experience

Westlake University Hangzhou, China
Visiting Research Intern, Inception3D, led by Prof. Anpei Chen *May–Oct 2026*
Investigated physics-constrained deformable dynamics, from recurrent state evolution to explicit MPM with graph-based residual correction; compared state representations, calibration, and correction interfaces for multi-step prediction. Authored a survey of 200+ sources on world-model validity and decision utility.

Tsinghua University, Institute for AI Industry Research (AIR) Beijing, China
Visiting Research Intern, DISCOVER Lab *Jan–Feb 2026*
Developed StereoHO-based human-to-robot handover with intent analysis, grasp planning around human-held regions, and stable transfer informed by object appearance and contents; implemented hand-eye calibration and robot execution.

Zhejiang University, Robotic Micro/Nano Manipulation Lab Hangzhou, China
Undergraduate Research Assistant, advised by Prof. Mingchuan Zhou *Feb 2025–present*
Developed hybrid physics–neural models of current-to-field mapping and actuation-to-motion response for magnetic microrobots; investigated Dreamer-based 3D navigation and PPO-based planar navigation with Transformer history encoding, contrastive context learning, and FiLM policy conditioning.

Zhejiang University, State Key Laboratory of CAD&CG Hangzhou, China
Undergraduate Research Assistant, advised by Prof. Ruizhen Hu *Oct 2025–Jun 2026*
Implemented dynamic street-scene reconstruction with COLMAP/FreeTimeGS and egocentric instance reconstruction with EFM3D/ShapeR; evaluated image-to-3D generation and articulated-object modeling.

Zhejiang University, Agile Robotic Tele-systems (ARTs) Lab Hangzhou, China
Research Assistant, advised by Prof. Gaofeng Li *Oct 2024–present*
Developed real-time stress-field reconstruction for layered-cast soft-tissue models, connecting sparse pressure measurements with curved-surface tactile sensing; optimized sensor layouts and used FEBio finite-element priors with Gaussian-mixture conditioning and posterior reweighting under uncertain material and loading conditions.

Publications

Zhenghe Guo. "ObjectTwin: Measuring the Asset Correspondence Gap for Capture-Anchored Digital Twins." Submitted; expanded manuscript in preparation. [Project page] [arXiv]

Fanghao Wang*, **Zhenghe Guo***, Yihao Hu*, et al. "Spatio-Temporal Transformer Meta-RL for Robust Magnetic Microrobot Navigation in Cluttered Environments." Submitted.

Zhenghe Guo. "Sample-Efficient Residual Dynamics for Contact-Mismatched Deformable Simulation." Submitted. [Project page] [Report]

Zhenghe Guo. "Beyond State Prediction: Learning Action Consequences in Visual Physical Systems." Submitted. [Project page] [Report]

X. Yang, **Z. Guo**, S. Liang, Z. Yang, C. Zhu, and Z. Zhang. "Rate Maximization for UAV-assisted ISAC System with Fluid Antennas." *2025 IEEE/CIC International Conference on Communications in China Workshops*, pp. 1–5, 2025. [Paper]

Zhenghe Guo et al. "Action Identification and Validation Design for Prediction with Missing Covariates." Submitted to *Journal of Statistical Planning and Inference*. Manuscript available upon request.

*Equal contribution.

Patents

Xiaoyan Chen, Xinyi Fang, Gaofeng Li, Qianming Chen, Zhichao Fan, **Zhenghe Guo**, Peng Deng, Misi Si, Rui Zhang. "Orthodontic Periodontal Ligament Stress Measurement Device and Stress Field Reconstruction Method." Chinese invention patent application. *Pending.*

Gaofeng Li, Zhichao Fan, Xiaoyan Chen, **Zhenghe Guo**, Xinyi Fang, Peng Cheng, Jiming Chen. "Method for Forming and Assembling a Simulated Periodontal Ligament Layer and Associated Testing Device." Chinese invention patent application. *Pending.*

Gaofeng Li, **Zhenghe Guo** (first student inventor), Xiaoyan Chen, Zhichao Fan, Xinyi Fang, Peng Cheng, Jiming Chen. "Physical Field Reconstruction Method and Device." Chinese invention patent application. *Pending.*

Selected Projects

DynaMo: Contact-Implicit Dynamic Motion Optimization and Energy-Aware Control for Humanoid Choreography [Problem] [Paper] [Project page] 2025

- Designed a dynamics-aware framework combining contact scheduling, ZMP-aware centroidal control, whole-body coordination, and electrical-energy analysis; Asia and Pacific Mathematical Contest in Modeling (APMCM) National First Prize (top 2%).

ZJUDancer: Vision & Localization for KidSize RoboCup Soccer [Team] 2025–2026

- Developed and debugged ball, field, and robot perception, self-localization, and odometry integration for autonomous humanoid soccer; National First Prize and Runner-up at RoboCup China Open 2026.

QUBO-Based Hybrid LDPC Decoding for Coherent Optical Optimization 2025
[Problem] [Paper] [Project page]

- Engineered an LDPC simulation and decoding prototype with belief-propagation baselines, QUBO construction, adaptive penalties, partitioned solving, and BER/FER/runtime evaluation.

RoboARM – Vision-Guided Gomoku Manipulation [Code] 2025

- The RoboARM system autonomously plays Gomoku using YOLO/ArUco board perception, Negamax move selection, ROS/HTTP communication, and a calibrated six-axis arm with a pneumatic end effector.

TurtleBot Autonomous Navigation [Project page] 2025

- The TurtleBot stack integrates Nav2 with a custom ROS RRT* planner, using goal-biased sampling, KD-tree collision checks, obstacle inflation, rewiring, and real-robot path execution.

Selected Awards

Robotics

- RoboCup China Open, Humanoid League (Small) – National First Prize, Runner-up, Team ZJUDancer, 2026.

Communication and Innovation

- Quantum Information Technology and Application Innovation Contest – National First Prize, Runner-up, 2025.
- China International College Students' Innovation Competition – Gold Medal, 2026.

Mathematical Modeling

- Asia and Pacific Mathematical Contest in Modeling (APMCM) – National First Prize (top 2%), 2025.
- DigitalCup Undergraduate Mathematical Contest in Modeling – National First Prize in Mathematics and Non-Mathematics groups, 2025.
- National Undergraduate Mathematics Competition, Non-Mathematics Category A – Second Prize, 2024.

Scholarship

- Xiangyang Scholarship, First Class (RMB 10,000; sole annual recipient in the college), 2024–2025.

Technical Skills

Programming	Python, C/C++, MATLAB, R, JavaScript/TypeScript, Git, Docker, Linux
Robotics	ROS, OpenCV, PyTorch, CUDA, calibration, kinematics and dynamics, path planning, embedded systems, Arduino/STM32/Raspberry Pi
Perception and ML	3D reconstruction, SfM, visual odometry, object detection, sensor fusion, reinforcement learning, optimization

Languages

Chinese (native) English (TOEFL 110)

Last updated October 2026. Detailed project reports, demos, and videos: https://styokuok.github.io/main/